Qiskit Fall Fest 2026

Challenge 1: Predict the Output

Problem Statement - A quantum circuit can produce different measurement outcomes depending on the gates applied to a qubit.
Your task is to predict the output of simple quantum circuits before executing them and then compare your prediction with the actual measurement results.

Your Task - Using Qiskit:

1. Create a single-qubit quantum circuit.
2. Prepare the qubit in the initial state $|0\rangle$.
3. Construct and investigate the following circuits:
    a. Circuit A: Apply an X gate and measure.
    b. Circuit B: Apply two X gates and measure.
    c. Circuit C: Apply an H gate and measure.
4. Before executing each circuit, predict the expected measurement outcome.
5. Execute each circuit for 100 shots.
6. Record the measurement results.
7. Compare your predictions with the actual results.
8. Display the results using a histogram.

Think Before You Run - Before executing the circuits, predict: What will happen after applying one X gate? What will happen after applying two X gates? What do you expect after applying an H gate? Will Circuit C always produce the same measurement result?

Design Constraints - Your circuit must:

- Use exactly 1 qubit.
- Include measurement.
- Use only the gates introduced in the workshop.
- Be created and executed using Qiskit.
- Use 100 shots for each experiment.

Submission - Your notebook must include:

- The three quantum circuits.
- Your prediction for each circuit.
- Measurement results for 100 shots.
- Histograms of the results.
- A short comparison between your predictions and the actual results.